ITS206 Software Construction and Design

Assessment C – Library Management System

Student Name

Suyog Basukala

Student ID

S2500054

Group Number

Group 4

Contents

1. Introduction 2

2. System Design 2

2.1 System Architecture 2

2.2 Class Design 2

3. Implementation 4

3.1 Concepts of Object-Oriented Programming 4

3.2 Data Structures 4

3.3 Premium Membership Extension 5

3.4 Due Date and Unavailability Checking 5

4. Testing 5

4.1 Unit Testing 5

5. Verification and Validation 9

6. Development Process Evidence 9

7. Challenges Faced 10

8. Conclusion 10

References (APA 7) 10

1. Introduction

The main objective of this project was to develop a Library Management System based on the concepts of Java programming and object-oriented programming. It was developed for efficient management of books, members, and borrowing processes by librarians.

In the application, users can perform functions such as adding, modifying, deleting, and searching books. In addition to adding and modifying members, borrowing functionality enables members to borrow books and return them. Furthermore, the project provides support for premium membership functionality as per the requirements of Group 4. Premium members have the ability to borrow up to six books, whereas normal members are restricted to borrowing three books.

The project shows how to utilize the concepts of object-oriented programming such as inheritance, abstraction, encapsulation, collections, validation, and error handling.

2. System Design

2.1 System Architecture

The application follows a simple object-oriented architecture consisting of six classes:

1. Book
2. Member (Abstract Class)
3. RegularMember
4. PremiumMember
5. Library
6. LibraryManagementSystem

The Library class acts as the central controller responsible for managing books and members.

2.2 Class Design

Book Class

Responsibilities:

- Store book information
- Manage borrowing status
- Track availability dates

Attributes:

- isbn

- title
- author
- borrowed
- availableDate

Member Abstract Class

Responsibilities:

- Storing book details
- Managing borrowing details
- Tracking available dates

Attributes:

- memberId
- name
- borrowedCount
- currentLoans
- borrowingHistory

RegularMember Class

Inherits from Member.

Borrow limit:

3 books

PremiumMember Class

Inherits from Member.

Borrow limit:

6 books

Library Class

Responsibilities:

- Managing books
- Managing members
- Borrowing and returning books
- Searching for books
- Keeping records of loans and borrowing histories

LibraryManagementSystem Class

Responsibilities:

- Menu display
- User input reception
- Function call for Library tasks

---

3. Implementation

3.1 Concepts of Object-Oriented Programming

Encapsulation

Private members are implemented in the Book class using getters and setters.

Sample:

Book details are prevented from being altered directly.

Inheritance

Two members inherit from the abstract Member class.

RegularMember

PremiumMember

This helps to save time by reducing maintenance tasks.

Abstraction

Abstract class is defined as follows:

Sample:

public abstract int getBorrowLimit();

Polymorphism

RegularMember and PremiumMember can be referred to as Member.

Sample:

Member member = new PremiumMember("M002", "Premium User");

This determines the appropriate borrow limit.

3.2 Data Structures

Use of ArrayList collections is made throughout the application.

Examples:

ArrayList<books>

ArrayList<members>

ArrayList<currentLoans>

ArrayList<borrowingHistory>

ArrayList was chosen due to their dynamic memory allocation and search capabilities.

3.3 Premium Membership Extension

A requirement from Group 4 was for Premium Memberships.

Memberships:

Normal Membership:

Borrowing limit of 3 books

Premium Membership:

Borrowing limit of 6 books

The system checks the borrowing limit prior to lending any books.

This fulfils the requirement of personalisation extension.

3.4 Due Date and Unavailability Checking

Whenever a book is borrowed:

availableDate = LocalDate.now().plusDays(14)

Calculation of the due date is made by the system.

If the book is not available, users are informed when the book will be available.

4. Testing

4.1 Unit Testing

Test Case 1

Objective:

Add a new book.

Input:

ISBN: B004

Title: Database Systems

Author: Connolly

Expected Result:

Book added successfully.

Actual Result:

Book added successfully.

Status:

Pass

```
Enter choice: 8
Enter ISBN: B004
Enter title: Database Systems
Enter author: Connolly
Book added successfully.

===== LIBRARY MANAGEMENT SYSTEM =====
1. Show Books
2. Show Members
3. Borrow Book
4. Return Book
5. Search Book
6. View Current Loans
7. View Borrowing History
8. Add Book
9. Update Book
10. Remove Book
11. Register Member
12. Update Member
0. Exit
Enter choice: 1
B001 | Java Basics | James Gosling | Borrowed: false | Available: 2026-06-07
B002 | OOP Concepts | Robert Martin | Borrowed: false | Available: 2026-06-07
B003 | Data Structures | Mark Allen | Borrowed: false | Available: 2026-06-07
B004 | Database Systems | Connolly | Borrowed: false | Available: 2026-06-07
```

Test Case 2

Objective:

Search for a book.

Input:

Java

Expected Result:

Java Basics displayed.

Actual Result:

Java Basics displayed.

Status:

Pass

```
Enter choice: 5
Enter title, author or ISBN: Java
B001 | Java Basics | James Gosling | Borrowed: false | Available: 2026-06-07
```

Test Case 3

Objective:

Borrow a book.

Input:

Member ID: M001

ISBN: B001

Expected Result:

Book borrowed successfully.

Actual Result:

Book borrowed successfully.

Status:

Pass

```
Enter choice: 3
Enter Member ID: M001
Enter ISBN: B001
Book borrowed successfully.
Due date: 2026-06-21
```

Test Case 4

Objective:

Borrow more books than allowed.

Input:

Borrow fourth book using regular member.

Expected Result:

Borrow limit reached.

Actual Result:

Borrow limit reached.

Status:

Pass

```
Enter choice: 3
Enter Member ID: M001
Enter ISBN: B004
Borrow limit reached.
```

Test Case 5

Objective:

Return a book.

Input:

ISBN: B001

Expected Result:

Book returned successfully.

Actual Result:

Book returned successfully.

Status:

Pass

```
0. Exit
Enter choice: 4
Enter Member ID: M001
Enter ISBN: B001
Book returned successfully.
```

Test Case 6

Objective:

View borrowing history.

Expected Result:

List of previously borrowed books displayed.

Actual Result:

History displayed correctly.

Status:

Pass

```
Enter Member ID: M001
Java Basics borrowed on 2026-06-07
OOP Concepts borrowed on 2026-06-07
Data Structures borrowed on 2026-06-07
```

5. Verification and Validation

Verification was carried out by matching the features implemented with those specified in the assessment specification.

Requirement	**Implemented**
Add Book	Yes
Update Book	Yes
Remove Book	Yes
Search Book	Yes
Register Member	Yes
Update Member	Yes
Borrow Book	Yes
Return Book	Yes
Due Dates	Yes
Current Loans	Yes
Borrowing History	Yes
Premium Membership	Yes
Availability Tracking	Yes

All required features as stated in the assessment specification have been implemented.

6. Development Process Evidence

Development was done using Git and GitHub.

The project was developed in phases using numerous commits.

Some of the commits involved were:

1. Initial library management system
2. Added member management functionality
3. Implemented borrowing feature
4. Added search functionality
5. Added return book functionality
6. Implemented premium membership feature
7. Added due date and availability tracking

8. Added interactive menu system
9. Added borrow limit checking
10. Added borrowing history and current loans
11. Added book management features
12. Added member registration and update features
13. Added input validation and error handling
14. Refactored project into final OOP design

Git history shows the progression of the software during development

7. Challenges Faced

There were various issues faced during the development process.

The first issue was dealing with a system capable of supporting both ordinary and premium users and ensuring code reuse. Inheritance and abstraction were used to address this problem.

The second issue was keeping track of the borrowing history as well as current borrowings. ArrayLists were separately used to keep track of both active and historical information.

Another problem was dealing with invalid user input to prevent runtime errors. Validation techniques were employed to check for user input validity.

8. Conclusion

Indeed, this project has successfully created a full-fledged Library Management System using Java and Object Oriented Programming concepts.

This software application enables the user to manage books, members, borrowing process, date of return, borrowing history, and premium membership. With the help of object-oriented programming concepts like abstraction, inheritance, encapsulation, and polymorphism, this application has been made more maintainable and scalable.

In addition, this project is an example of the implementation of software development concepts.

References (APA 7)

1. Deitel, P., & Deitel, H. (2023). *Java How to Program* (12th ed.). Pearson.
2. Schildt, H. (2021). *Java: The Complete Reference* (12th ed.). McGraw-Hill.
3. Oracle Corporation. (n.d.). Java documentation. https://docs.oracle.com/java/

4. Bloch, J. (2018). *Effective Java* (3rd ed.). Addison-Wesley.
5. Eckel, B. (2006). *Thinking in Java* (4th ed.). Prentice Hall.